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From Conversations to Knowledge: Enriching Movie Datasets with Knowledge Graph

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***Abstract*—Knowledge graphs are invaluable assets for facilitating the dissemination of structured and semantically enriched data to both humans and machines while ensuring its accuracy and reliability. However, it has been observed that a direct approach of generating knowledge graph from existing benchmark data sources such as ReDial, DBPedia, etc. fails to highlight many crucial relationships between entities. In this work we proposed a strategy to enrich a knowledge graph by resolving multiple entity relationships generated from the ReDial movie dataset. First the missing entities and their corresponding relationships were identified. Subsequently, the publicly available large TMDb dataset is leveraged to align additional information such as genre, director, actor, writer, and year as nodes. Bidirectional links are then introduced to semantically enrich the knowledge graph. Validating a knowledge graph is considered as another important step. In this direction, we extract triplets in (movie, relationship, genre) format from the knowledge graph. The triplets are validated using IMDb which is a vast repository of movie details, and the Gemini Generative AI model. We compute accuracy of the extracted triplets and the results indicate improvement in the knowledge graph generation strategy and the IMDb as an efficient validator for our approach.**

***Index Terms*—Knowledge Graph, TMDb, ReDial, Named Entity Recognition, IMDb, Gemini, Fact Validation**

I. Introduction

Knowledge graph (KG) serves as graphical representations of accumulated human knowledge, provide invaluable support to various artificial intelligence tasks, including question answering, social recommendation, logical reasoning, text generation, code analysis, etc. Inspired by their efficacy, recent scholarly attention has shifted towards enhancing KG models. The aim is to produce superior entity and relation representations within KGs, thereby elevating overall performance [1]. A KG is a structured representation of knowledge that uses a graph-based data model. It organizes information into nodes and edges, where nodes represent entities or concepts, and edges denote relationships between them. There are a number of rule-based and machine learning-based methods for scalable relationship extraction is available in literature [2]. For instance the traditional machine learning-based approaches such as Naive Bayes, random forests, and the modern deep learning transformers such as DistilBERT [3], PubMedBERT [4], T5 [5], and SciFive-based models [6]. KGs can provide a framework for semantic knowledge representation from given textual information. However, in order to build a KG, it is necessary to extract knowledge as relationships between entities and normalize both entities and relationship types [2].

KGs enable efficient storage, retrieval, and inference of interconnected knowledge, facilitating advanced search, analysis, and reasoning tasks. Fact validation in a KG is a task to determine whether a given fact (subject, predicate, object) should appear in the KG. The verified facts can be used for multiple tasks such as; incomplete KGs refining, KGs violation detection, improving the quality of knowledge search, and multiple KGs integration [7].

The primary objective of our research is to validate factual statements presented in triplet form such as (subject-predicate-object) relationship. e.g. Subject-Movie-Inception, Predicate-Relationship-has_genre, Object-Genre-Science Fiction. So, the factual statement for above example is Inception has genre Science Fiction. This validation process serves as a foundational step for various applications, including recommendation systems, question answering, and other related tasks.

In summary, our main contributions of this work are as follow,

- Integration of ReDial with TMDb to include the crucial release year, genre, director, actor etc attribute, addressing the limitation of missing temporal information in the KG.
- Preprocessing and alignment techniques to resolve anomalies such as missing entities and spelling variations, enhancing the accuracy and reliability of entity representation in the KG.

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II. Related Work

This literature survey explores recent advancements in KG generation and validation, offering insights into methodologies and challenges in this field.

Huaman et al. [8] highlight the importance of KG validation, discussing challenges like reproducibility inconsistency and limited adaptability. They advocate for a robust validation framework with triple verbalization and result visualization to enhance research credibility.

Similarly, [9] focused on fact validation in KGs, particularly for the Semantic Web Challenge. They utilized RDF2Vec embedding features and machine learning techniques, training a Random Forest classifier on both positive and negative statements to predict fact veracity. Real-world datasets like DrugBank were used for model validation, resulting in high prediction accuracy.

Milovsevic et al. [2] extract biomedical relationships using rule-based and machine learning methods, favoring Transformer-based models for superior performance. Perevalov et al. [10] propose a query system generating KGs from natural language queries, enhancing question answering.

zhang et al. [11] a framework for error-aware KG embedding to address common errors in KG construction. AEKE integrates entity attributes and consists of three main components: a KG representation learning model with confidence scores, a triple confidence learning module leveraging structural and attribute information, and a joint adaptive training scheme. This approach aims to mitigate noise impact on embedding vectors and enhance the overall reliability of KGs.

Geng et al. [12] Addressing KGC challenges, the proposed Relational Message Passing Network (RMPN) infers target triples' likelihood directly from relations, bypassing entity features. It iteratively extracts subgraphs to represent relation relationships, propagates features, and computes scores, enabling unseen relation inference and optional ontological schema augmentation.

The key hypothesis of our work is KG triplets validations over generated KG to better stimulate the strength of KG.

Ren et al. [13] Utilizing the ReDial dataset in conjunction with DBpedia, augmented the dataset by associating movie titles with director, writer, actor, producers and genre information, enhancing its richness. However, it's worth noting that the mapping for the year attribute is not considered. Similarly, Sarkar et al. [14] explored combination of ReDial and DBpedia for entity mapping and employs a generative model for entity linking. The dataset comprises of 6,924 movies and 4,765 mentioned entities. Notably, 824 movies and 121 mentioned entities could not be linked to DBpedia, resulting in isolated nodes in the graph.

From the approaches mentioned above, we observe the presence of isolated nodes in the KG poses a challenge as they do not contribute to establishing relationships between entity-attribute pairs, leading to a lack of additional information. This becomes particularly crucial as such information is later utilized for knowledge-based tasks, especially in conversational recommendation systems. Therefore, addressing the issue of isolated nodes is essential to enhance the effectiveness and utility of the KG for various applications.

For instance, the ReDial dataset when combined with DBpedia fails to incorporate the release year attribute, which is crucial for various search operations on the generated KG. To address this gap, we integrate ReDial with TMDb, which is a comprehensive movie database that includes the year attribute along with the other essential attributes such as genre, year, etc.. Moreover, during the mapping of movie titles with attributes, we identify errors such as missing entities, which are resolved through preprocessing to refine the ReDial dataset. Subsequently, we align the results from TMDb with the preprocessed movie-attribute mapping, using an intersection to match and unify entities with spelling variations. This approach reduces the complexity of KG nodes and enhances the accuracy of entity representation.

Thus our study addresses the challenge of error-prone entity mapping in KGs by implementing preprocessing techniques to streamline KG construction. This simplification of the KG creation process aims to enhance accuracy and reliability. Furthermore, we leverage Resource Discription Framework (RDF) Query for KG Visualization to ensure the accurate representation of triplets. To bolster the robustness of our approach, we validate our results using a test set from TMDb and cross-reference with IMDb and Gemini, thus ensuring the effectiveness and applicability of our methodology for KG generation.

III. Proposed Methodology

In this section, we delineate our proposed approach for enriching the dataset with supplementary movie information. A comprehensive overview of the methodology is depicted in 1.

A. Dataset Preparation and Preprocessing

In this work, we explore the benchmark ReDial (Recommendation Dialogue) movie dataset as the corpus of conversation and their corresponding recommendations [15]. Existing literature [16], [17], [13], [18], [19] have widely used ReDial as it comprises of a huge volume of around 11,000 conversations, movies recommended for each conversation.

However, the dataset cannot be directly used for generating KG from the conversations. Data preprocessing was necessary because the ReDial dataset had a format that contained numerous variables that were not relevant to the KG generation.

For enriching the movie dataset by exploiting KG, we require `conversationId` and a list containing the utterances of both the user and system

Our primary objective in preprocessing was to establish a more efficient indexing mechanism to expedite retrieval of entire conversations, reformat the data for easier application of Named Entity Recognition (NER), and improve the overall readability of the data.

The benchmarked ReDial is split into test and train sets for validating a typical conversational recommender system

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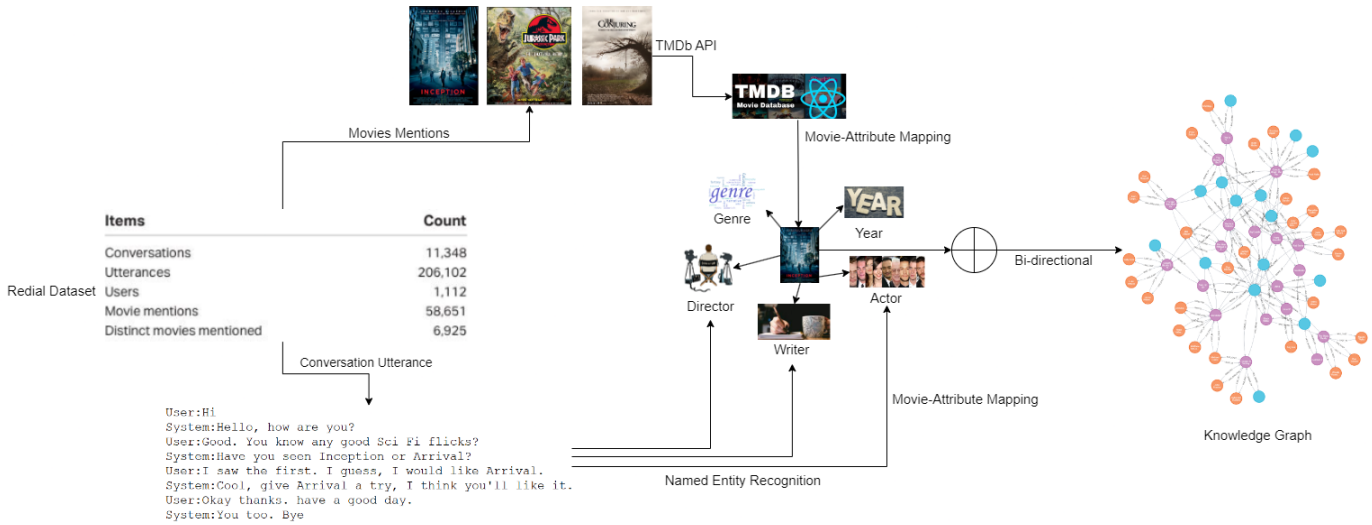

Fig. 1: An Overview of the proposed methodology.

models [15]. However, to generate the initial KG we merged them to create a single corpus of 11,348 conversations. This amalgamation lays the groundwork for subsequent KG construction.

First, we identify movie titles within the conversations from the `movieMentions` variable in the dataset that maps movie IDs to Movie names. Structuring these entities such as conv_id: {Movies} ensures precise tagging of each conversation with relevant movie details.

The subsequent task involves compiling a comprehensive list of unique movies mentioned across the dataset. Employing a systematic approach, movie titles are collated and deduplicated, ensuring the accuracy and completeness of the dataset.

Concurrently, textual content extraction from conversations is carried out. This process involves extracting all textual content from each conversation and organizing it into a structured format, typically a list of messages. Such an organization facilitates further analysis and processing of conversational data within the KG framework.

B. Named Entity Recogntion

NER is deemed crucial for extracting key entities from conversational text due to its ability to identify and classify specific types of entities. The NER step extracts key entities from the conversation text. Initially, the BERT model [20] is exploited to extract entities such as locations (LOC), organizations (ORG), and persons (PER) from the text. For instance, in the conversation, "I really enjoyed Leonardo DiCaprio's performance in Inception," the named entity "Leonardo DiCaprio" would be recognized as a person (PER). By utilizing pre-trained models, entities of interest, like persons (PER) in this example, are efficiently extracted, laying the groundwork for further analysis and entity linking within our KG.

C. Data Enrichment from TMDb

Our primary objective is to augment the working dataset with supplementary information about the movie, such as genres, actors, directors, and writers. TMDb 5000 movie dataset [21] is a popular online database that provides information about movies, TV shows, and celebrities, including details such as plot summaries, cast and crew information, release dates, and ratings. Leveraging the extensive resources provided by TMDb, we accessed a wealth of metadata associated with movies, including comprehensive details about their production and personnel. This involved querying the TMDb API [1] to retrieve relevant information such as genre classifications and the names of individuals involved in various roles within the film industry.

For example, consider the movie "Inception." While the conversational text may not explicitly mention its genre, leveraging TMDb allows us to retrieve this information. Querying the TMDb API for "Inception" returns its genre as "Science Fiction," providing valuable additional context to enrich the working dataset.

Furthermore, to ensure coherence and relevance to the working dataset, we performed an intersection operation between the TMDb data and the entities extracted through NER approach. Specifically, we filtered the TMDb personnel data to include only those individuals who intersected with the previously extracted person entities from our conversations. By aligning the TMDb personnel with the entities identified within the working dataset, we maintained consistency and relevance. This approach enabled an enriched KG with high-quality and contextually relevant information.

D. Entity-Attribute Mapping

Semantic relations are established between movies and personnel within the KG. This involves defining several key steps to define semantic relations, ensure bidirectional mapping, and organize data into a structured format for efficient processing.

First, semantic relation is defined by considering four random types; 'has director', 'has writer' 'has cast', and 'has

[1] https://api.themoviedb.org/3/search/movie

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genre'. Throughout the entirety of our approach, we treat the term 'cast' as synonymous with 'actor.'. These relations served as the backbone of our KG by facilitating the linkage between movies and the associated personnel involved in their production. By explicitly defining these relations, we aimed to capture the inherent connections between different entities within the film industry, thereby enriching the semantic depth of the proposed KG.

Second, bidirectional mapping of the above relations in the generated KG is performed. This approach enables easy traversal of the generated graph in both directions. Also enabled nuanced queries and analysis of the generated graph elements. For instance, not only could we ascertain which movies a particular director worked on, but also which directors were associated with a given movie.

Finally, we organized the extracted data into a structured format using a common JSONL (JSON Lines) file. This file encapsulated essential information about movies, including their associated genres, cast members, directors, and writers. By structuring the data in this standardized format, we facilitated ease of access and streamlined the mapping process to establish relations between entities.

E. Generation of the KG

The generation of the KG involved two key steps; Node Creation and Relationship Mapping.

Initially, the movies, personnel, and genres extracted from both the working dataset and existing TMDb dataset is considered as nodes for the graph. Each entity was instantiated as a node within the graph, with unique identifiers and properties that captured relevant attributes such as movie titles, director, writer, actor, and genre classifications. By creating nodes for these entities, we established the foundational elements of our KG, enabling the representation of diverse entities and their interconnections.

Subsequently, to establish relationships between these nodes based on the predefined semantic relations and data from the JSONL file. Leveraging the defined relations such as 'has director', 'has writer' 'has cast', and 'has genre', we forged connections between relevant nodes within the graph.

By mapping these relationships within the graph, we constructed a network of interconnected nodes that encapsulated the semantic associations between movies, personnel and genres.

After following the above methodology, the final KG is generated which contains 7515 nodes comprising Movie titles, Directors, Writers, Actors, Genres and Years and 47,250 relationships of the types DIRECTED, HAS_DIRECTOR, WROTE, HAS_WRITER, CAST_IN, HAS_CAST and GENRE_OF, HAS_GENRE. The distribution can be seen in Fig 2. The relationships are structured to capture the bidirectional properties between nodes as well. Below is a Cypher query that retrieves a part of the KG where nodes labeled as a "Movie" have all the relationships and returns the other nodes that satisfies the relationship.

```
MATCH (m:Movie)-[:CAST_IN|DIRECTED|GENRE\_OF
|HAS_CAST|HAS_DIRECTOR|HAS_GENRE
|HAS_WRITER|WROTE]->(related)
RETURN m, related LIMIT 50
```

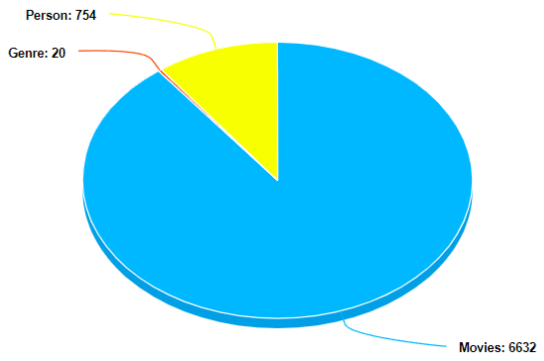

Fig. 2: Distribution of node types in the KG.

In Figure 3 the KG is depicted based on the execution of the query mentioned above.

F. Experimental Setup

In this section, we detail the hardware devices, tools, and publicly available resources employed for validating the proposed work.

The experiments were conducted on a personal computer featuring an AMD Ryzen 5 4600H processor with 6 cores and 12 threads, coupled with 16 GB DDR4 RAM. Additionally, it is equipped with a 4GB NVIDIA GeForce GTX 1650 Ti Graphics card for GPU-accelerated computations. The primary programming languages and frameworks utilized include Python, TensorFlow, and PyTorch, supported by essential software libraries such as NumPy, Pandas, and Py2neo. For development, tools like Google Colab, Visual Studio Code, and Neo4j were employed to facilitate efficient coding and experimentation. The ReDial dataset [15] is used as the baseline for implementing this work. Whereas, the TMDb [21] is used for enriching the KG with entity relationships based on ReDial. Additionally, IMDb [22] is used one publicly available repository for validating the movie related triplets from the KG.

G. Validation

Validating the generated KG is an important step. In this regard, various publicly available resources are exploited. Firstly, all the available triplets are extracted from the KG.

For example, consider the following triplet extracted from the KG:

$$\text{Movie: "Inception"} \xrightarrow{\text{has_genre}} \text{Genre: "Science Fiction"}$$

This triplet indicates that the movie "Inception" is associated with the genre "Science Fiction," providing a clear relationship between the movie entity and its corresponding genre.

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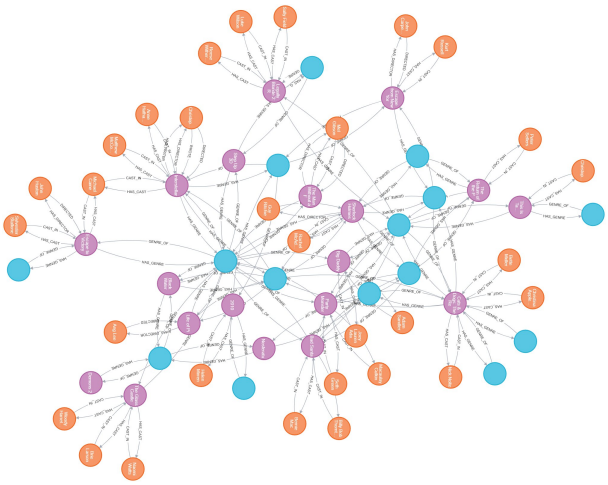

Fig. 3: Knowledge graph representation derived from the executed query.

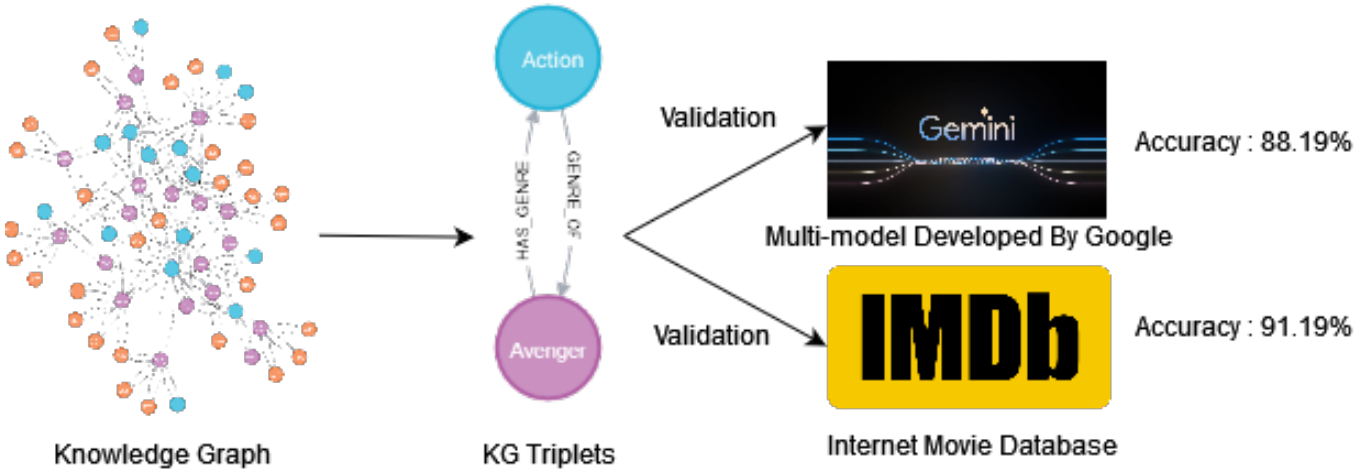

Fig. 4: Overview of the KG triplet validation.

Now these extracted triplets were subsequently validated using a publicly available dataset and a generative artificial intelligence model. We chose IMDb (Internet Movie Database) [2] as the public dataset due to its extensive coverage of movies, including detailed information on cast, crew, plot summaries, and user ratings. IMDb is a widely recognized online database that holds a vast repository of movie-related information. It is extensively used by millions of users worldwide for its reliable and comprehensive data on films, television shows, and entertainment content. IMDb includes detailed entries on movie titles, genres, production details, cast and crew, plot summaries, user reviews, and ratings. This extensive and structured movie-related data makes IMDb an ideal resource for validating the accuracy and completeness of our KG triplets. By leveraging IMDb's data, we can ensure that the relationships and entities in our KG are both accurate and well-supported by a trusted source. Further, the Gemini generative artificial intelligence model by Google [23] was also employed for the validation task of the generated KG.

Choosing Internet Movie Database (IMDb) and Gemini for validating KG triplets is a strategic decision. IMDb offers a vast repository of structured movie-related data, making it ideal for validating movie-related triplets in the KG. Additionally, IMDb's widespread usage and reliability ensures accuracy of the approach. On the other hand, Gemini, as an AI-driven chatbot developed by Google, provides a complementary approach to validation. Its natural language understanding capabilities allow for nuanced validation beyond simple data matching. To systematically validate the KG, we employed the following step-by-step approach:

1) From the created Knowledge Graph (KG), extract all the triples and store them.
2) Pass all the triples extracted through the knowledge sources, i.e., IMDb and Gemini.
3) Assign a score of 1 if the triple represents correct information when compared to a single knowledge source, and do the same for the second knowledge source.

[2] $https://www.imdb.com/$

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Assign a score of 0 if the triple represents incorrect information.

4) Plug in the values in the accuracy equation provided in the section below.

An overview of the KG validation process is shown in Fig. 4

H. Results and Discussion

The primary evaluation metric used in this analysis is accuracy. Accuracy gives us a measure of how close our measurements are to the real value. KG accuracy can be defined as the percentage of triples in the KG being correct Gao et el [24]. Hence, we have adopted the Eq. (1) that describes accuracy as:

$$\text{Accuracy} = \left(\frac{\text{Number of Correctly Validated Triples}}{\text{Total Number of Triples}}\right) \times 100\% \tag{1}$$

The results in Table I depict the performance of the proposed KG compared to two other knowledge sources: IMDb and the Gemini generative AI model.

Interestingly, during the validation of our proposed KG generation methodology using IMDb and Gemini, we observed that IMDb provided a greater enrichment of supplementary information for entities. While IMDb is a comprehensive movie database, Gemini is a multimodal large language model developed by Google. As demonstrated, our generated KG achieves an accuracy of 88.19% when compared to Gemini, while exhibiting a slightly higher accuracy of 91.19% when compared to IMDb for KG triplets validation.

For producing a more consistent, accurate, and useful KG, the error sources need to be identified and tackled. We observed certain limitations which may effect the performance of a KG,

1) Nodes were compromised while fetching the triplets from the KG. These mostly include Unicode errors where some foreign characters not recognized by the system are replaced by their Unicode values. Some examples of these would be:
   - Actual Name: ¡Three Amigos! (1986)
     Corrupted Name: `\u00A1Three Amigos! (1986)`
   - Actual Name: Déjà Vu (2006)
     Corrupted Name: `D\u00E9j\u00E0 Vu (2006)`
2) Communication errors encountered during the utilization of IMDb and Gemini APIs as knowledge sources could be attributed to the high volume of queries, resulting in some requests not being properly executed and potentially leading to missed hits. As a consequence, a certain degree of miscellaneous errors may have occurred due to this reason.

IV. Conclusion

In conclusion, our study presents a robust methodology for enhancing movie datasets through KG augmentation. By leveraging data from the ReDial dataset and supplementing it with information from TMDb, we constructed a comprehensive KG encompassing movie titles, directors, writers, actors, genres and years interconnected through semantic relationships. Through meticulous preprocessing, named entity recognition, and data enrichment techniques, we ensured the coherence and relevance of the generated KG. Validation using IMDb and Gemini showcased high accuracy rates, affirming the utility of our approach. Furthermore, by representing the KG in RDF, we adhere to a standardized format for semantic data representation, enhancing interoperability and facilitating integration with existing KGs and semantic web applications. This study's findings have implications for various applications, including conversational recommendation systems, where enriched movie datasets can provide personalized and contextually relevant recommendations to users. Despite encountering challenges such as data preprocessing issues and communication errors, our study demonstrates the potential of KGs to enrich and contextualize movie datasets for conversational recommendation systems and other applications. Through Knowledge graph triplet validation process, we ensure adaptability and effectiveness, achieving promising accuracy levels and validating the efficacy of our KG generation methodology.

TABLE I: Accuracy comparison between IMDb and Gemini using Proposed KG Generation Approach on ReDial Dataset

Knowledge Source	Accuracy
IMDb	91.19%
Gemini	88.19%

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